

Deepak Mallampati

APPLIED AI / FORWARD DEPLOYED ENGINEER

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
US-based, F-1 OPT - open to UK/SF relocation w/ sponsorship

PROFESSIONAL SUMMARY

AI/ML engineer with 4+ years building the data and experiment-automation infrastructure behind production AI - custom RAG pipelines over messy real-world corpora, model evaluation harnesses that catch regressions before deploy, and fault-tolerant orchestration. Comfortable in the model-in-the-loop / human-in-the-loop layer: curating and cleaning training and evaluation datasets, automating experiments, and turning research concepts into production-ready infrastructure. Strong Python, data wrangling, and ML-workflow fundamentals, with a research background (peer-review-ready paper) and a track record of shipping reliable systems for regulated, high-stakes domains.

CORE COMPETENCIES

Data Pipelines & Dataset Curation • Experiment Automation • Model Evaluation (RAGAS, BERTScore) • Python Scripting & Automation • RAG & Retrieval • Human-in-the-Loop / Model-in-the-Loop • MLOps & Observability • Privacy-Preserving ML

Data & Pipelines: Python, Pandas, NumPy, Spark, Databricks, Kafka | PostgreSQL, MongoDB, Redis | data cleaning, normalization, augmentation, labeling | data lineage & governance

ML & Evaluation: PyTorch, TensorFlow | feature engineering, dataset curation | RAGAS, BERTScore, custom domain metrics | drift detection, A/B testing | fine-tuning (LoRA, QLoRA, PEFT)

GenAI & Retrieval: LangChain/LangGraph | RAG architecture | multi-agent / conductor-subagent orchestration | FAISS, Pinecone, Weaviate | cross-encoder re-ranking

Infra & MLOps: Docker, Kubernetes, Terraform | AWS, Azure, GCP | CI/CD (Jenkins, Azure DevOps) | MLflow, Weights & Biases | batch orchestration, retry logic, monitoring

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, improving pipeline reliability and reducing failed production runs by **60%**.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency and accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Engineered high-performance **RAG pipelines** over large-scale document corpora, combining dense retrieval with cross-encoder re-ranking to lift relevance and reduce hallucination on domain-specific queries.
- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing objectives into context-scoped subagents - reducing token consumption by **42%** while sustaining task accuracy.
- Designed **privacy-preserving data workflows** for sensitive datasets (k-anonymity, l-diversity, differential privacy) enabling compliant analytics with zero PII exposure.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Built a **privacy-preserving ML pipeline** on a clinical dataset (k-anonymity $k=3$, l-diversity $l=2$, Laplace DP across four ϵ levels), reducing re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance.
- Designed a composite **privacy-utility scoring framework** to benchmark six anonymization configurations and identify the optimal operating point.
- Curated a **361-example instruction dataset** and fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 for controllable long-form generation.
- Authored a peer-review-ready **IEEE-format conference paper** documenting methodology, experiments, and results.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Developed AI agents and backend services for an internal AI platform across **25+ agents**, integrating with production data pipelines and observability tooling.
- Built **12 APIs and microservices**, reducing p95 latency from 1.5s to **800ms** and supporting **100,000+ requests daily** across 20+ internal teams.
- Evaluated GPT-4o, Claude Sonnet, and Gemini, improving task success rates by **27%** on internal evaluation datasets.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Executed end-to-end **data preprocessing** - cleaning, normalization, and transformation of raw sensor and operational datasets - to produce model-ready training pipelines.
- Built supervised regression/classification models for equipment-failure prediction and anomaly detection; fine-tuned BERT/RoBERTa for NER (**89% F1**).

PROJECTS

career-ops

An autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - built and operated end-to-end as a single engineer.

Privacy-Preserving Clinical ML Pipeline

A framework combining k-anonymity, l-diversity, and differential privacy that quantifies exactly how much privacy each technique buys for how little accuracy cost.

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites - a productivity tool taken from concept to public release solo.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0